

Data Sources

This document describes the external data sources used to construct the analysis dataset (floodopioid_demo.dta). The provided dataset is pre-merged and ready for analysis; these instructions are for reference and full reproducibility from raw sources.

1. FEMA Presidential Disaster Declarations (Flood Events)

- **Source:** Federal Emergency Management Agency (FEMA)
- **URL:** <https://www.fema.gov/openfema-data-page/disaster-declarations-summaries-v2>
- **Variables:** fefl, fefl_f11h, fefl_f11h_1st_year, fefl_f11h_2nd_year, all_fefl_f11h_1st_year, never_fefl_f11h, fema_allother, fema_allother_last1, fema_allother_last2, fema_allother_last3
- **Description:** County-level FEMA Presidential Disaster Declarations (PDDs). Incident types labeled as “Flood,” “Severe Storm,” and “Hurricane” are classified as flood-related disasters. Non-flood FEMA declarations (e.g., wildfires, earthquakes) are used as controls.
- **Access:** Public, no registration required

2. NOAA Storm Events Database

- **Source:** National Centers for Environmental Information (NCEI), NOAA
- **URL:** <https://www.ncdc.noaa.gov/stormevents/>
- **Variables:** fl1d, fl1h, fl1d_1st_year, fl1d_2nd_year, all_fl1d_1st_year, never_fl1d, F_D_*, L_D_*
- **Description:** County-level storm events with associated fatalities, injuries, and property damage. For flood identification, the following event types are classified as flood-related: “Coastal Flood,” “Flash Flood,” “Flood,” “Debris Flow,” “Heavy Rain,” “Hurricane,” “Lakeshore Flood,” “Strong Wind,” “Thunderstorm Wind,” “Tropical Depression,” “Tropical Storm,” and “Tornado.” Data are used starting in 1996, as the NOAA Storm Events Database underwent significant changes in both data collection process and event categorization prior to 1996.
- **Access:** Public

3. CDC NCHS Restricted-Use Mortality Data (Opioid Overdose Deaths)

- **Source:** National Center for Health Statistics (NCHS), CDC
- **URL (restricted):** <https://www.cdc.gov/nchs/nvss/nvss-restricted-data.htm>
- **URL (public CDC WONDER):** <https://wonder.cdc.gov/mcd.html>
- **CDC guidelines for identifying opioid deaths:** <https://www.cdc.gov/nchs/nvss/vsrr/drug-overdose-data.htm>
- **Variables:** death_opioid_*, death_alldrug_*, death_pre_*
- **Description:** The longitudinal opioid-related overdose death data, at the county-by-year level, were constructed using the NCHS’s Restricted-Use Vital Statistics of the Detailed Multiple Cause of Death Data. This micro-level dataset includes a single record for each death, with information on causes of death, the deceased’s county of residence, time of death, and demographic characteristics including age, gender, race, and education. Drug overdose deaths

are identified based on underlying cause-of-death codes X40–X44 (unintentional), X60–X64 (suicide), X85 (homicide), and Y10–Y14 (undetermined). Opioid-related involvement is identified based on contributing cause-of-death codes T40.0 (opium), T40.1 (heroin), T40.2 (natural opioid analgesics), T40.3 (methadone), T40.4 (synthetic opioid analgesics other than methadone), and T40.6 (other and unspecified narcotics).

- **Access:** Restricted-use data requires a research proposal review and approval by the NCHS. Publicly available suppressed data are accessible through CDC WONDER.

4. CDC Bridged-Race Population Estimates

- **Source:** U.S. Census Bureau in collaboration with the National Center for Health Statistics (NCHS)
- **URL:** <https://wonder.cdc.gov/bridged-race-population.html>
- **Variables:** `pop_allpop`, `pop_male`, `pop_female`, `pop_white`, `pop_nonwhite`, `pop_less25`, `pop_2544`, `pop_4564`, `pop_over65`
- **Description:** Annual county-level bridged-race population estimates by age, sex, and race, 1999–2017. Produced by the U.S. Census Bureau and NCHS to provide consistent population denominators for computing vital statistics rates.
- **Access:** Public

5. American Community Survey (ACS) 5-Year Estimates (Education)

- **Source:** U.S. Census Bureau
- **URL:** <https://www.census.gov/programs-surveys/acs>
- **Variables:** `edupct_1eh`, `edupct_hh`, `pop_1eh`, `pop_hh`
- **Description:** County-level population by educational attainment (less than high school vs. bachelor's or higher). Each 5-year estimate was assigned to its midpoint year to create an annual panel; for example, the 2006–2010 ACS estimates were assigned to 2008. The earliest ACS 5-year estimates used were the 2006–2010 release, so county-level population breakdowns by educational attainment from the ACS are available from 2008 onward. For 1999–2007, county-level educational attainment was drawn from the 2000 Decennial Census.
- **Access:** Public

6. Census Population Estimates Program (Net Migration)

- **Source:** U.S. Census Bureau, Population Estimates Program
- **URL:** <https://www.census.gov/programs-surveys/popest/technical-documentation/methodology.html>
- **Variables:** `rnetmig`, `domesticmig`, `rdomesticmig`, `popest`
- **Description:** County-level annual net migration estimates, which combine multiple data sources to estimate migration. The net migration rate is defined as net domestic migration plus net international migration per 1,000 residents, where positive values indicate net in-migration and negative values indicate net out-migration.
- **Access:** Public

7. Bureau of Labor Statistics (Unemployment Rate)

- **Source:** U.S. Bureau of Labor Statistics, Local Area Unemployment Statistics
- **URL:** <https://www.bls.gov/lau/tables.htm#cntyaa>
- **Variables:** `unemployrate`
- **Description:** County-level annual unemployment rate from the Local Area Unemployment Statistics (LAUS) program.
- **Access:** Public

8. Small Area Income and Poverty Estimates (SAIPE) — Poverty and Median Household Income

- **Source:** U.S. Census Bureau, Small Area Income and Poverty Estimates (SAIPE) Program
- **URL:** <https://www.census.gov/programs-surveys/saipe/data/datasets.html>
- **Variables:** `povertypct`, `medianhhi`
- **Description:** County-level annual estimates of poverty rate and median household income from the SAIPE Program.
- **Access:** Public

9. Appalachian Regional Commission (ARC) County Designations

- **Source:** Appalachian Regional Commission
- **URL:** <https://www.arc.gov/county-economic-status-and-distressed-areas/>
- **Variables:** `econcate2002`, `econcate2003`, `econcate2004`, `ruralregion`, `appasubregion`, `pctnation2000_medianhhi`, `pctnation2000_unemploy`, `pctnation2000_poverty`
- **Description:** ARC county economic status classifications (Distressed, At-Risk, Transitional, Competitive, Attainment) and rural/urban designation within the Appalachian region. 420 Appalachian counties identified.
- **Access:** Public

10. County Shapefiles (TIGER/Line)

- **Source:** U.S. Census Bureau, TIGER/Line Shapefiles
- **URL:** <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>
- **Files:** `US_Continental_Counties.*` (county boundaries), `tl_2023_us_state.*` (state boundaries)
- **Description:** Geographic boundary files for mapping. County shapefile covers continental U.S. Appalachian counties; state shapefile used for context boundaries.
- **Access:** Public